

Wilfred Bates

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Education

MSc Scientific Computing & Data Science, University of Bristol: Sep 2025 – Aug 2026 (Due)

Interdisciplinary degree focused on machine learning, large-scale data processing, software engineering for scientific research, and high-performance computing.

- **Group Project:** Developing analysis software for neutron scattering experiments using Bayesian statistical curve fitting to extract physical parameters from highly complex experimental data incorporating bayesian/ akaki model selection.

BSc Natural Sciences (First Class Honours, 77%), University of Exeter: Sep 2022 – Jun 2025

Awarded *Outstanding BSc Graduate (top of cohort)* and two Dean's Commendations (top 5%). Quantitative pathway with emphasis on computational, mathematical, and data-driven science, with a foundation in all life sciences.

- **Dissertation:** Performed complete scRNA-seq workflow including all preprocessing, QC, normalisation, feature selection and dimensionality reduction; applied modern tools including Dynamo, Scanpy and AnnData for lineage inference, velocity analysis and trajectory reconstruction. Evaluated tool limitations and robustness of biological interpretation.
- **Group Project (Dataset Complexity & CNN Optimisation):** Investigated how dataset characteristics influence CNN performance; developed experimental pipeline, managed task allocation and reviewed code quality, taking on coordination and mentoring responsibilities within the team.
- **Mathematical Modelling in Biology and Medicine:** Constructed and performed extensive analysis of numerical models of complex biological/clinical systems with MATLAB. Utilises stability analysis and mathematical stability investigations.
- **Medical Data Science (Applied Module):** Extensive introduction to R with a self directed project using data science and statistical methods to investigate medical data.
- **Artificial Intelligence & Applications (Applied Module):** Applied computer science module with extensive hands on work in building and testing neural networks for classification tasks as well as foundational theory in traditional ML methods.

A Levels & GCSEs

A Levels: Maths, Physics, Computer Science (AAA)

GCSE: 11 GCSEs; Grade 9 in English Language and Mathematics

Work Experience

Data Science Intern, Infinite Code Works

Oct 2025 – Present

- Assisted in forecasting future product demand using recurrent neural networks (DeepAR) on AWS.
- Prepared and transformed data into stationary time series to enable accurate forecasting of component demand.
- Presented analysis results and forecasts to clients and internal team members.

Collaborator, Low Europe

Feb 2025 – Aug 2025

- Supported projects across energy, planning, SMEs, and digital competition.
- Developed and presented AI solution analyses to senior management (benefits, feasibility, timelines, costs).
- Automated workflows using ChatGPT API and Airtable to enhance outreach and content management.
- Contributed to a pilot multi-agent system for automated EU Commission tender processing using NLP.

Head Counsellor, Camp Beckett

Jun 2023 – Aug 2023

- Led a team of 8 boys (aged 13-15) and 2 staff members, planning and executing 2 daily activities and classes, This required enormous energy and creativity and provided invaluable lessons in leadership and team building.

Skills

Technical: 8+ years Python(Matplotlib, Numpy, Pandas, TensorFlow, scikitlearn), R(Tidyverse), LaTeX, Git, Command Line, MATLAB, C/C++, AI API Usage

Attributes: Enthusiasm, Communication, Teamwork, Problem-solving, Presenting, Adaptability, Independence

Extra Curricular

- Vice President, Natural Sciences Society: Organised popular events including wellbeing socials, park runs, and extracurricular lectures.
- Volunteer, Raleigh International (Nepal, 10 weeks) – Contributed to building infrastructure and delivering awareness sessions in rural communities, developing strong teamwork skills through challenging and rewarding experiences.
- Charity Running – Raised alone over £2,000 for various charities through long-distance events.
- Head of House & Committee Member (School) – Coordinated external speakers for audiences up to 200; introduced talks and led Q&A sessions. Across these roles I strengthened leadership, teamwork, communication, and organisational skills.